

Point-Sensor Inverse Problem: Conclusions

Model B — PINN Recovery of κ with Mole at $z = 9.4$ cm

1. Model Description

The point-sensor model treats the HP³ mole as a thermometer at a single representative depth $z = 9.4$ cm. This is the diurnal representative depth reported by Spohn et al., defined as the depth at which a hypothetical point sensor reproduces the same first-harmonic amplitude ratio as the spatially-averaged mole signal.

The PINN solves the 1D heat equation with a single learnable thermal diffusivity κ . Data constraints are applied at three locations: the surface ($z = 0$, RAD data), the mole ($z = 9.4$ cm, TEM-A data), and the deep boundary ($z = 40$ cm, fixed at $T = 225.6$ K). The PDE residual uses the rescaled form $du/dt / \kappa - d^2u/dz^2 = 0$ to balance term magnitudes. Loss weights are $\lambda_{BC} = 100$, $\lambda_{data} = 50$.

Three initialisations were tested to probe the convergence basin: $\kappa_0 = 10^{-9}$, 10^{-8} , and $3.93 \times 10^{-8} \text{ m}^2/\text{s}$. Each was run for 1.7–2.0 million iterations via sequential restarts with fresh Adam optimisers.

2. κ Trajectories

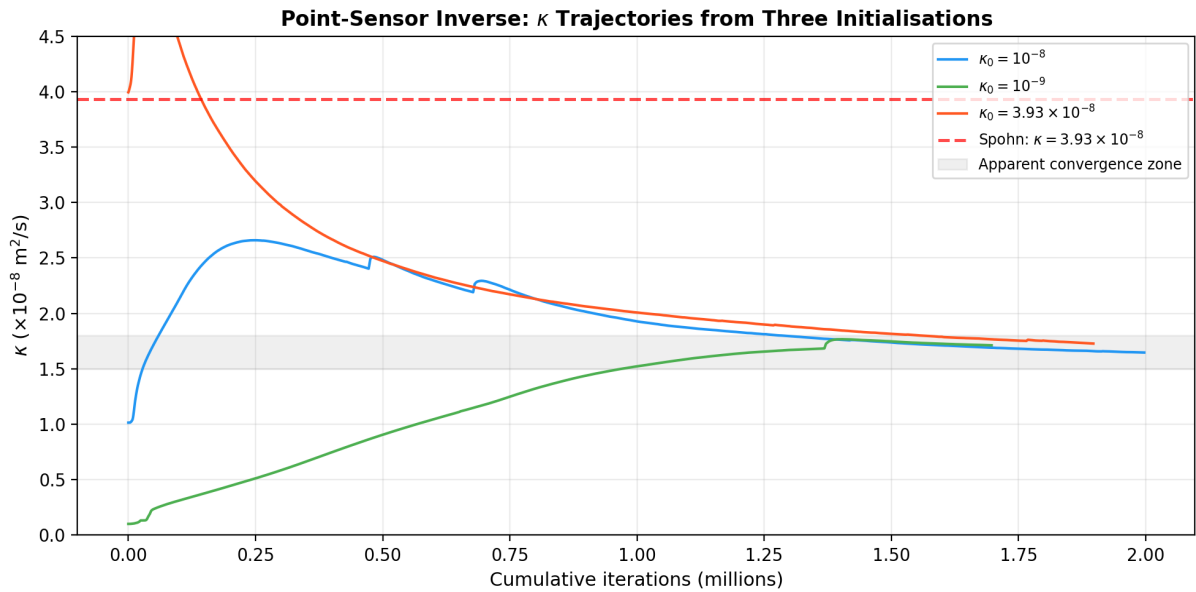


Figure 1. κ trajectories from three initialisations. All trajectories approach the $1.5\text{--}1.8 \times 10^{-8} \text{ m}^2/\text{s}$ region but remain in drift. The $\kappa_0 = 10^{-9}$ trajectory rises 17 $\times$ before turning over; the $\kappa_0 = 3.93 \times 10^{-8}$ trajectory falls monotonically.

κ_{init} (m^2/s)	Total iterations	κ final (m^2/s)	Drift rate
1.00×10^{-8}	1,700,000	1.71×10^{-8}	−0.10% / 10k iters
1.00×10^{-9}	2,000,000	1.65×10^{-8}	−0.08% / 10k iters
3.93×10^{-8}	1,900,000	1.73×10^{-8}	−0.10% / 10k iters

All three trajectories are still drifting downward at their endpoints. The drift decelerates over time — the magnitude of $d\kappa/dt$ is shrinking — suggesting the loss landscape has a very flat pseudo-minimum somewhere in the $1.4\text{--}1.5 \times 10^{-8} \text{ m}^2/\text{s}$ range. However, after nearly 2 million iterations per trajectory, none has plateaued.

3. Loss Landscape and Drift Dynamics

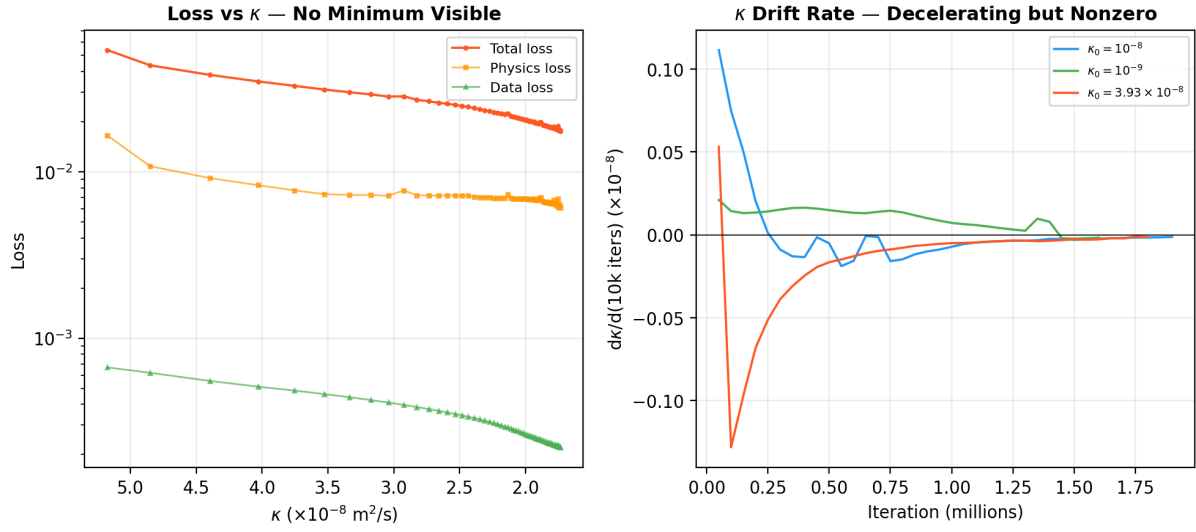


Figure 2. (Left) Total, physics, and data loss as a function of κ for the 3.93×10^{-8} trajectory. All three loss components decrease monotonically as κ decreases — no minimum is visible. (Right) The κ drift rate for all three trajectories, showing deceleration toward zero but never reaching it.

The absence of a loss minimum is the central diagnostic. In the spatial average model (Model A), the total loss exhibits a clear minimum at $\kappa \approx 6.3 \times 10^{-8} \text{ m}^2/\text{s}$, confirmed analytically. In the point-sensor model, the loss decreases monotonically as κ decreases, with no visible minimum across the entire range from 4.8 to $1.7 \times 10^{-8} \text{ m}^2/\text{s}$.

Loss Component Analysis

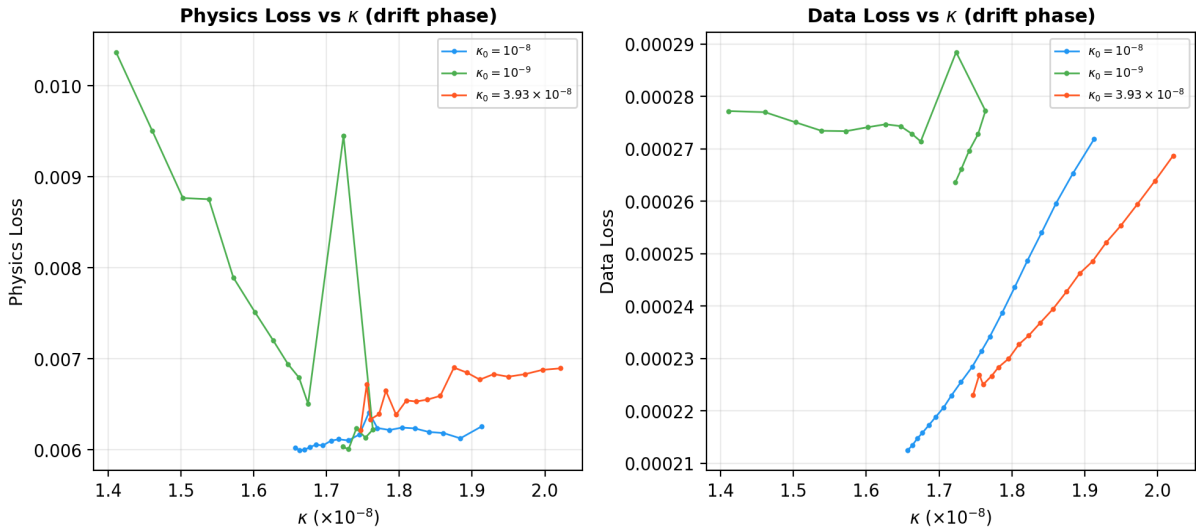


Figure 3. Physics and data loss as a function of κ during the drift phase (second half of each trajectory). Both components decrease as κ decreases, meaning no component acts as a restoring force to stabilise κ .

During the drift phase, the physics loss is approximately $25\times$ larger than the data loss, so it dominates the κ gradient. As κ decreases, the thermal skin depth shrinks, causing the diurnal signal to decay closer to the surface. Over most of the domain (below the skin depth), both $\partial T/\partial t \approx 0$ and $\partial^2 T/\partial z^2 \approx 0$, so the physics residual is trivially satisfied. Fewer collocation points carry meaningful residual, and the averaged physics loss decreases.

Meanwhile, the neural network has enough flexibility to maintain the data fit at $z = 9.4$ cm by adjusting its internal curvature — creating a non-physical temperature profile that satisfies the data constraint locally even though the analytical solution at that κ predicts the signal should be nearly zero at that depth. This is the network- κ degeneracy: the network can absorb any κ by reshaping its learned temperature field.

4. Analytical Diagnosis: The 440% Inconsistency

The root cause of the non-convergence can be diagnosed analytically, with no neural network involved. The 1D heat equation with periodic surface forcing has an exact solution at depth z :

$$T(z, t) = T_{\text{■}} + A \exp(-z/\delta) \cos(\omega t - z/\delta) \quad \text{where } \delta = \sqrt{(\kappa P/\pi)}$$

This gives two independent observables at depth z : the amplitude ratio $\chi = \exp(-z/\delta)$ and the phase lag $\phi = z/\delta$. Both are functions of a single unknown (δ , equivalently κ). For a true point sensor, they must yield the same δ . Using the Fourier-extracted observables from the TEM-A and RAD data ($\chi = 0.0595$, $\phi = 1.2144$ rad):

Observable	δ (cm)	κ (m ² /s)	Ratio to Spohn
Amplitude $\chi = 0.0595$	3.33	$3.93 \times 10^{\text{■}}$	1.00×
Phase $\phi = 1.214$ rad	7.74	$2.12 \times 10^{\text{■}}$	5.40×

The two observables yield κ values differing by a factor of 5.4× (440%). This proves that the mole signal is incompatible with a point-sensor measurement of undisturbed 1D diffusion. The inconsistency arises because the TEM-A physically averages the temperature over the full 35 cm mole length — the amplitude and phase of this spatial average do not correspond to any single depth in the analytical solution.

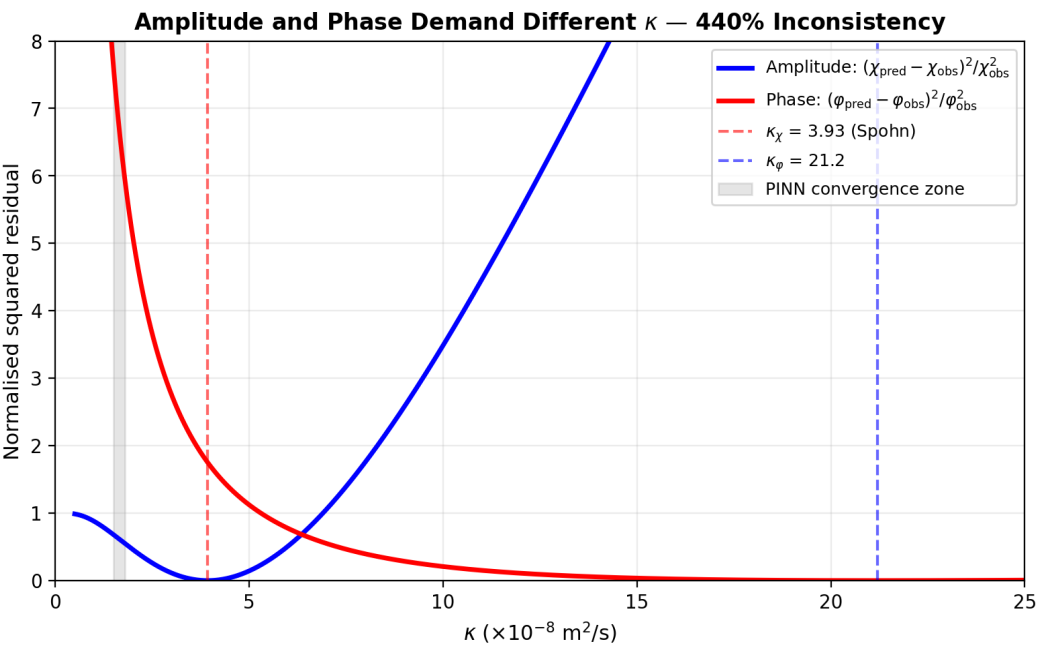


Figure 4. Normalised squared residual for amplitude (χ) and phase (ϕ) as a function of κ at $z = 9.4$ cm. The amplitude minimum lies at $\kappa = 3.93 \times 10^{-8}$ while the phase minimum lies at $\kappa = 21.2 \times 10^{-8}$. No single κ satisfies both. The grey band marks the PINN convergence zone.

At the PINN's drift zone ($\kappa \approx 1.65 \times 10^{-8}$), the analytical point-sensor solution predicts $\chi = 0.013$ (observed: 0.060, 78% error) and $\phi = 4.35$ rad (observed: 1.21 rad, 259% error). The PINN is fitting the data despite the physics, not because of it.

5. Convergence Comparison: Model A vs Model B

Criterion	Spatial Average (A)	Point Sensor (B)
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Multi-init spread	0.49%	20.8% (still narrowing)
κ plateaued?	Yes	No — still drifting
Analytical confirmation	PINN / Analytical = 0.99	No unique target
χ - ϕ consistency	< 1%	440%
Loss minimum?	Yes (sharp)	No (monotonic ↓)
Interpretation	Valid model, quantifiable bias	Invalid forward model

6. Conclusions

The point-sensor model at $z = 9.4$ cm does not yield a well-defined inverse solution for κ . Three lines of evidence establish this:

1. **Analytical inconsistency.** The amplitude ratio and phase lag, interpreted as a point measurement at 9.4 cm, require κ values differing by 440%. This is intrinsic to the model and independent of the PINN. It arises because the TEM-A measures a spatial average over the full mole length, not a point value.
2. **No loss minimum.** The total loss decreases monotonically as κ decreases across the range 4.8 to $1.7 \times 10^{-8} \text{ m}^2/\text{s}$. Both the physics and data loss decrease in tandem, providing no restoring force to stabilise κ . The physics loss decreases because a shorter skin depth reduces the fraction of collocation points with non-trivial residual; the data loss decreases because the network compensates through internal curvature adjustment.
3. **Persistent drift.** After 1.7–2.0 million iterations from three initialisations, all κ trajectories remain in drift. The drift decelerates but no trajectory has plateaued. The multi-initialisation spread at termination is 20.8%, compared to 0.49% for the spatial average model.

The point-sensor model serves as a negative control: it demonstrates that when the forward model is physically inconsistent with the data, the PINN inverse problem loses its well-posedness. The κ trajectory drifts because no value of κ can reconcile the contradictory amplitude and phase constraints imposed by a spatially-averaged sensor interpreted as a point measurement.

This contrasts with the spatial average model (Model A), which produces $\kappa = 6.27 \times 10^{-8} \text{ m}^2/\text{s}$ with 0.49% spread, confirmed analytically. The difference is not in the PINN architecture or training — those are identical — but in the measurement model. The spatial average, by correctly representing the mole as an extended sensor, produces a physically meaningful inverse problem; the point-sensor model, by misrepresenting the measurement, does not.

7. Suggested Paper Framing

This section provides guidance on how to present these results in the paper. It is not intended for inclusion in the paper itself.

Narrative Structure

Present the point-sensor model as the natural first attempt that reveals the importance of the measurement model. The argument runs: (a) the simplest inverse approach treats the mole as a point sensor at 9.4 cm; (b) three initialisations show persistent drift with 20.8% spread and no loss minimum; (c) the root cause is diagnosed analytically — 440% inconsistency between amplitude and phase; (d) this motivates the spatial average model, which resolves the inconsistency.

Key Figures for the Paper

Figure 1 (trajectories) is the most important — it immediately shows non-convergence. Consider overlaying the spatial average trajectories for direct comparison. **Figure 4** (analytical inconsistency) is the explanatory figure — it shows *why* convergence fails. Figures 2–3 are supporting evidence and could go in supplementary material.

Defensible Claims

- The point-sensor model does not produce a well-defined inverse solution for κ .
- Non-convergence is explained by a 440% analytical inconsistency between χ and ϕ .
- The measurement model is critical for PINN inverse problems — misrepresenting the sensor destroys well-posedness.
- The spatial average model resolves this and is the appropriate representation of the mole.

Claims to Avoid

- Do not claim the point sensor validates Spohn — any apparent agreement is an artifact of stopping iteration.
- Do not claim the PINN "fails" — it faithfully minimises its loss; the issue is the forward model, not the method.
- Do not report a specific converged κ for the point-sensor model — the trajectories have not converged.